

Responses Overview Active

Responses

10

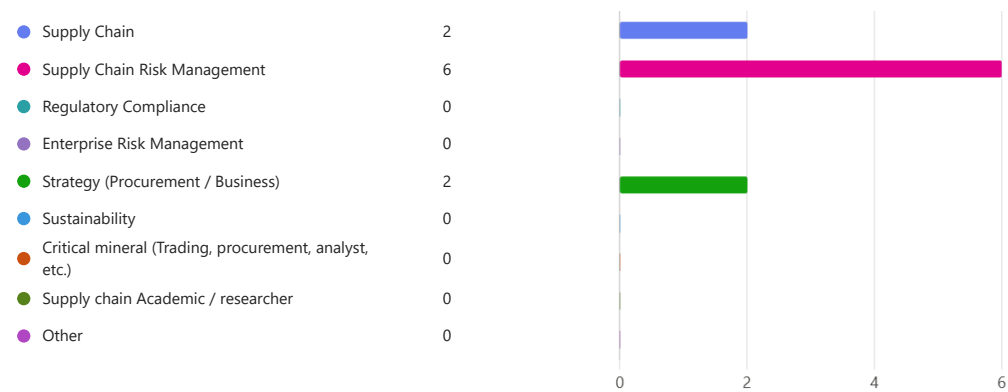
Average Time

387:59

Duration

67 Days

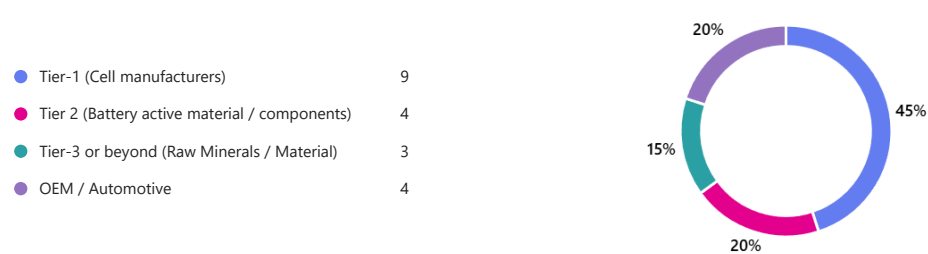
1. Which of the below field best describes your professional experience / expertise



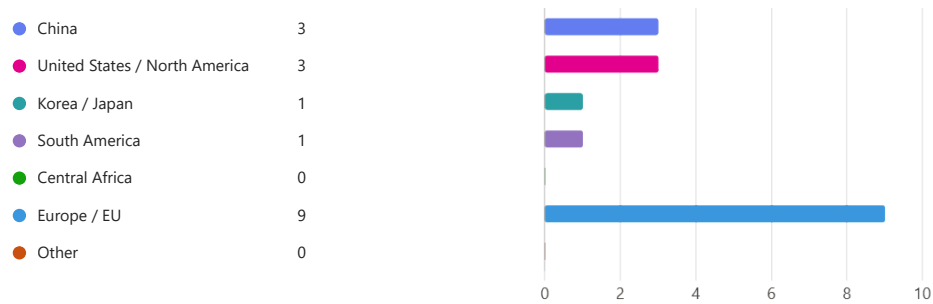
2. How many years of experience do you have in EV battery supply chains, critical mineral sourcing, or supply chain risk management ?



3. In which supply chain tier(s) do you have professional experience/s? (Select all that apply)

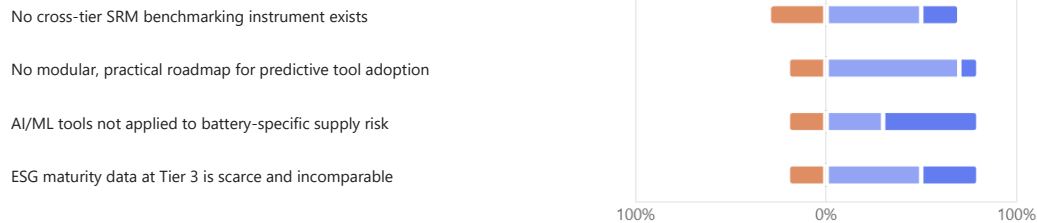


4. What is your primary region of expertise? (multiple choice)



5. The proposed research addresses this core problem: "No replicable, cross-tier instrument exists for benchmarking strategic risk management (SRM) maturity across Tier 1 cell manufacturers and Tier 3 raw material suppliers — and no evidence-grounded, modular roadmap exists for integrating predictive risk tools at different maturity levels in this sector."

Not a real gap Minor gap Significant gap Critical gap



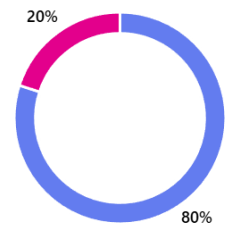
6. In case the answer for above question is **Not a Real Gap / Minor Gap**, please provide a comment or provide the study / instruments that addresses

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Responses

Latest Responses
"We are developign one with Ashsishs Engine. Also Prewave and EA are using AI el..."
...

7. To your knowledge, does any tool, standard, or published study **already address** cross-tier SRM benchmarking for battery supply chains in a replicable, comparable way?

Response	Count
No — nothing comparable exists	8
Partially — something similar exists: [Please describe briefly]	2
Yes — the following addresses it fully: [Please describe briefly]	0



8. In case the answer for above question is **Partially / Yes**, please provide provide the study / instruments that addresses

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Responses

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"Would have to be done by comparing individual approaches and layouts"
...

9. From your experience, which of the following **risk domains** are most underserved by current SRM practices in the battery sector? (*Rank 1 = most underserved, 5 = least underserved*)

Rank 1 Rank 2 Rank 3 Rank 4 Rank 5

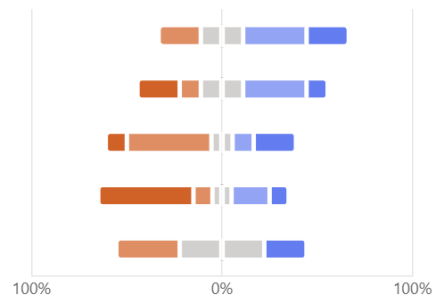
Supply disruption — concentration, geopolitical exposure

ESG incidents — human rights, artisanal mining, environmental

Regulatory compliance — EU Battery Regulation, CSRD, TCFD

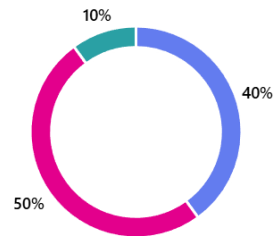
Predictive/digital tools — data quality, ML readiness

Financial risk — commodity price volatility, hedging



10. The proposed research focuses only on **Tier 1 cell manufacturers** and **Tier 3 raw material suppliers**, intentionally excluding Tier 2. Does this focus on the two most exposed tiers make sense to you?

- Yes — Tiers 1 and 3 are the right focus for this research 4
- Partially — Tier 2 should also be included because: 5
- No — the most important tier for this research is actually 1



11. In case the answer for above question is **Partially / No**, please provide the study / instruments that addresses

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"They are also a critical part of the supply chain and could pose a risk if not assess... "

"Because process chains are individual and have unique historical patterns that ca... "

...

2 respondents (33%) answered raw material for this question.

stream chemicals historical patterns critical step chemicals processing

chains are individual process chains supply chain

risk bottlenecks Tier raw material risk critical

Gulf War complete check material level SCs War for example

reason aluminum foil

12. The proposed research restricts its scope to two risk domains: **supply disruption** and **ESG incidents**. Which, if any, additional risk domains do you believe are impossible to exclude from a meaningful SRM benchmarking exercise?



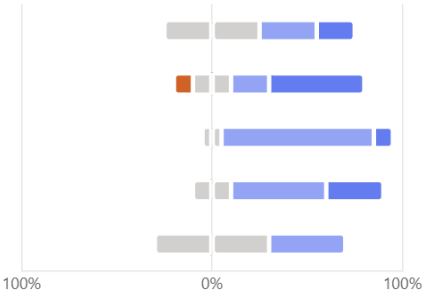
2 respondents (33%) answered supply for this question.



13. The thesis uses five dimensions to assess SRM maturity. Please rate each dimension on how well it captures a real, measurable aspect of SRM in the EV battery supply chain

Rating -1 Rating 2 Rating 3 Rating 4 Rating 5

- D1 — Risk Governance (board oversight, 3 Lines of Defence, risk appetite)
- D2 — Supply Disruption Management (concentration risk, BCP, dual sourcing)
- D3 — ESG Risk Management (supplier audits, human rights due diligence)
- D4 — Predictive & Digital Risk Tools (scenario analysis, ML, dashboards)
- D5 — Disclosure & Reporting Quality (TCFD, GRI, sub-tier data disclosure)



14. Are there any **critical SRM dimensions missing / to be added** from this framework that you believe should be included?

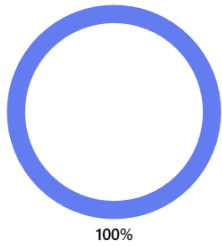


2 respondents (40%) answered supply chain for this question.



15. The thesis uses a **5-level maturity scale** (Initial → Emerging → Defined → Integrated → Optimized). Does this scale reflect the maturity progression you observe in industry practice?

- Yes — the five levels map well to observable industry behaviour 10
- Partially — the following level(s) are unclear in practice 0
- No — I would suggest a different scale 0



16. In case the answer for above question is **No / Partially**, please provide a response below

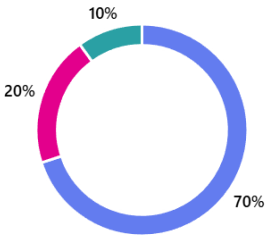
0 responses submitted

0
Responses



17. The thesis awards a **Level 4 (Integrated) score** only when a firm provides "**explicit, documented process description**" (Tier A evidence) in a public corporate disclosure. Does this evidence standard appropriately reflect what genuine Level 4 capability looks like — or does it risk systematically underrating firms whose best practices are proprietary and not publicly disclosed?

- Appropriate — public disclosure evidence is the right standard for a transparent benchmark 7
- Too strict — genuinely capable firms will be systematically underrated because 2
- Too lenient — public disclosures overstate actual capability because 1



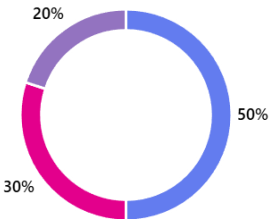
18. In case the answer for above question is **Too strict / Lenient**, please provide a response below

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...

19. The benchmark will be built using **publicly available corporate disclosures** (annual reports, sustainability reports, risk management reports) rather than internal firm data. In your experience, do corporate disclosures reliably represent a firm's **actual** SRM capability?

- Yes — what firms disclose publicly reflects their internal capability reasonably well 5
- No — disclosures consistently **overstate** capability, especially in: 3
- No — disclosures consistently **understate** capability, especially in: 0
- It depends — the disclosure-capability gap is largest at: 2



20. In case the answer for above question is **No / It depends**, please provide a response below

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21. For **Tier 3 raw material suppliers** (miners, trading companies), public disclosures are much thinner. What data sources do you consider the most reliable proxy for assessing Tier 3 SRM capability, in the absence of firm-level disclosures? *Examples: TE/Cobalt Congo reports, GBA Battery Passport pilots, RMI certifications, OECD Due Diligence audit results, journalist investigations*

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Responses

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"Incidence and news occurrences on the 5 dimensions"
...

1 respondents (20%) answered reportsGovernmental regulators for this question.

energy NGO no info
Incidence neverSocial media news occurrences BGR in Germany
oversight reportsGovernmental regulators insider knowledge
dimensions Audits and eyewitnesses mining entities associations
knowledge/interview Public representation representation of the company

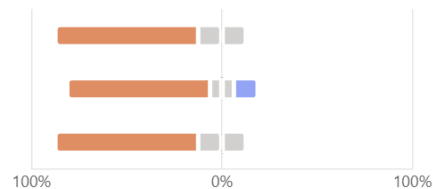
22. The research proposes to benchmark the following **Tier 1 cell manufacturers** as primary cases. Please indicate your view of each as a representative benchmark case:

● Good representative case ● Acceptable but not ideal ● Poor choice — reason below

CATL (China — market leader)

LG Energy Solution (South Korea — ESG-intensive)

Panasonic Energy (Japan — Toyota/Tesla dual-OEM partner)



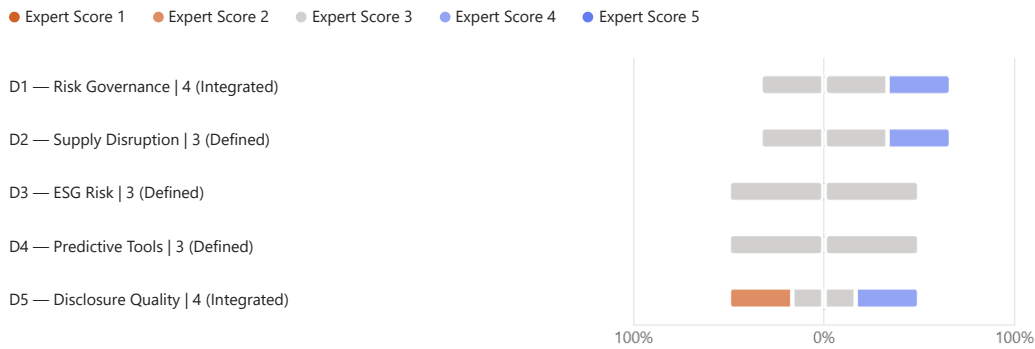
23. If any are options are marked **poor choices**, please explain below:

Also, Are there **Tier 1 cell manufacturers missing** from this list whose SRM practices you consider significantly different from CATL, LGE S, and Panasonic — and who should be included? (*Examples: Samsung SDI, BYD, ACC, SVOLT*)

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Responses

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24. CATL Assessment Score v/s Expert Opinion

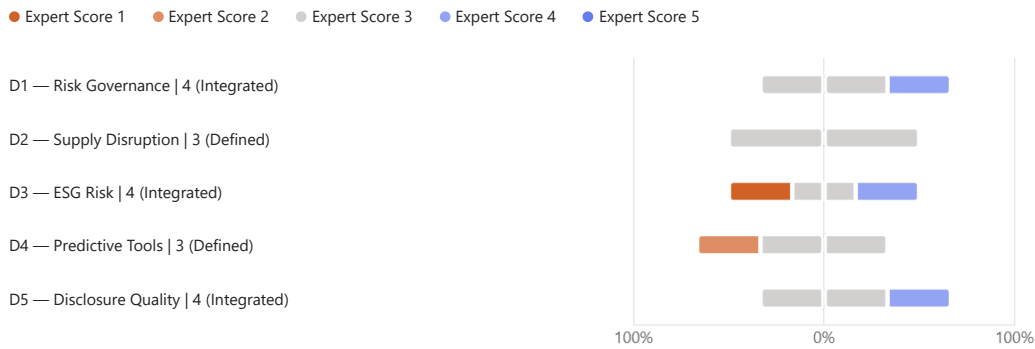


25. Provide comments on CATL Scoring (if required)

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Responses

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26. LG Assessment Score v/s Expert Opinion

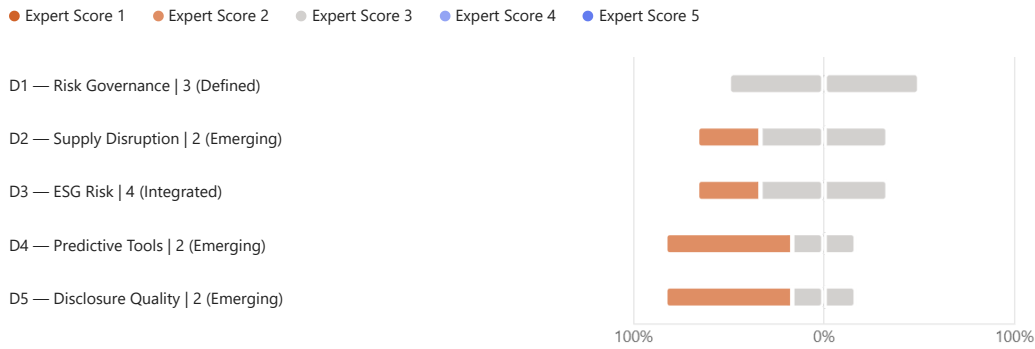


27. Provide comments on LG Scoring (if required)

2
Responses

Latest Responses
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28. Panasonic Assessment Score v/s Expert Opinion
(Note: Panasonic's lower scores partially reflect disclosure gaps, not necessarily lower capability)

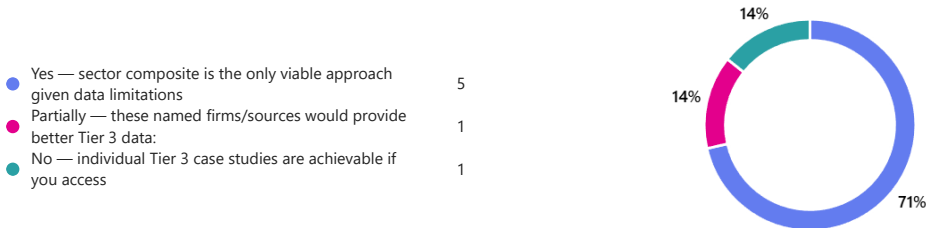


29. Provide comments on Panasonic Scoring (if required)

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Responses

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30. The research cannot benchmark **individual Tier 3 firms** due to data opacity and will instead use a **sector-level composite** (drawing on TE 2024 cobalt reports and GBA Battery Passport pilot findings). Does this approach capture the Tier 3 reality adequately?



31. Incase the answer for above question is **No / Partially**, please provide a response below

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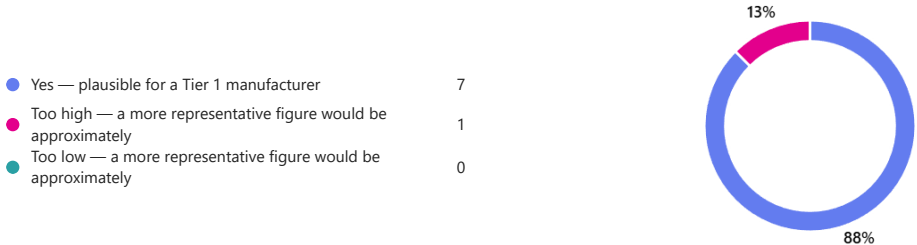
32. In your experience, are there **exceptions at Tier 3** — major industrial mining companies whose SRM maturity is significantly higher than artisanal and small-scale mining operations — that should be separated from the sector composite?

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Responses

Latest Responses
"Yes. The big miners should be more mature"
...

33. The thesis models annual critical material procurement spend at **500 Million €** for a representative Tier 1 cell manufacturer.

Is **annual critical (or Tier-3) material procurement spend** plausible as an industry average for Tier 1 cell manufacturer ?



34. In case the answer for above question is **Too high / Low**, please provide a response below in **Million €**

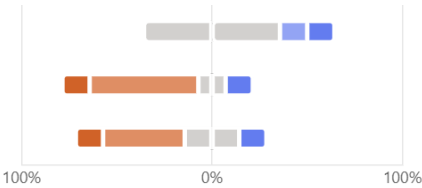
0
Responses



35. In your professional experience, what is the approximate probability that a Tier 1 cell manufacturer experiences **at least one significant supply disruption** per critical mineral in any given 12-month period?

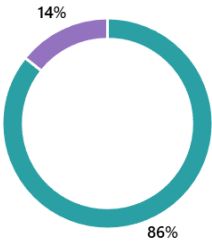
<5% 5-15% 15-30% 30-50% >50%

Cobalt (DRC-sourced)
Lithium (Lithium Triangle)
Nickel (Indonesia/ Philippines)



36. When a supply disruption occurs, what is a realistic **average duration** before material supply is restored or substituted?

Less than 2 weeks 0
2-6 weeks 0
Option 36-13 weeks (1-3 months) 6
13-26 weeks (3-6 months) 1
Over 6 months 0



37. For each week that a critical material supply is disrupted, what percentage of annual material procurement cost does a Tier 1 manufacturer typically absorb as operational loss (production downtime, emergency sourcing premium, contract penalties)?

Under 0.5% per week 1
0.5-1.5% per week 0
1.5-3% per week 4
3-5% per week 1
Over 5% per week 0



38. Based on your experience please provide range for annual material procurement cost

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Responses

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...

39. In your professional experience, how likely is a **material ESG compliance incident** at Tier 3 (e.g., child labour finding, environmental violation, conflict mineral misreporting) to occur within a 12-month period for a typical Tier 3 cobalt or lithium supplier?



40. Please provide an estimate for ESG Probability (Low risk - Base Case - Stress Case)

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Responses

Latest Responses
"Base Case"
...

41. When an ESG incident is discovered and confirmed, what is a realistic **total cost** to the Tier 1 cell manufacturer (regulatory penalty + reputational damage + remediation + sourcing disruption), considering EU Regulation 2023/1542 penalty structures?



42. Please provide an estimate for ESG Cost / Penalty ranges (Low - high Case)

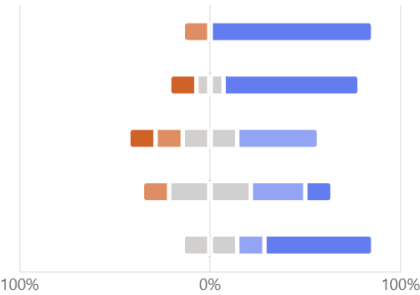
3
Responses

Latest Responses
...

43. The simulation will model three specific **supply disruption interventions**. Please rate each on how impactful and realistic it is for Tier 1 firms to implement:

1 2 3 4 5

- Dual-sourcing (multiple suppliers per mineral)
- Strategic safety stock (buffer inventory)
- ESG audit investment (reducing incident probability)
- Long-term offtake contracts with governance clauses
- Geographic diversification of sourcing



44. The simulation will model three specific **supply disruption interventions**. Please rate each on how impactful and realistic it is for Tier 1 firms to implement:

1 2 3 4 5

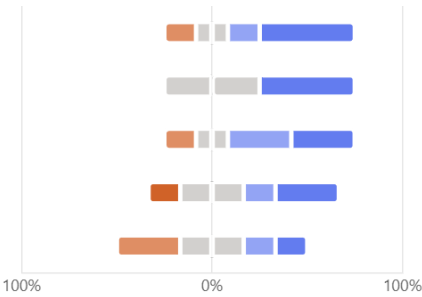
Dual-sourcing (multiple suppliers per mineral)

Strategic safety stock (buffer inventory)

ESG audit investment (reducing incident probability)

Long-term offtake contracts with governance clauses

Geographic diversification of sourcing



45. Are there **critical interventions missing** from this list that Tier 1 firms actually use to manage supply disruption or ESG incident risk?

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Responses

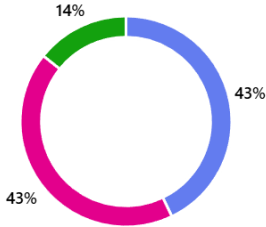
Latest Responses
"Technology Development/Roadmap towards reducing critical minerals/materials"
...

46. The research will develop a **three-stage modular adoption roadmap** for predictive risk tools:

- **Stage 1** — Structured analytical tools (risk heat maps, scenario matrices) — low cost, low data requirement
- **Stage 2** — Quantitative simulation (Monte Carlo modelling) — medium cost, medium data requirement
- **Stage 3** — Machine learning integration (AI-based risk monitoring) — high cost, high data requirement

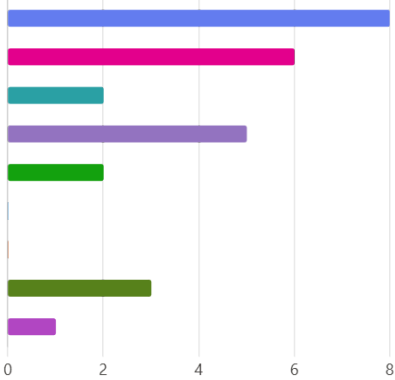
Does this three-stage progression match how you observe firms actually adopt risk analytics capabilities?

- Yes — this sequencing reflects typical capability development paths 3
- Partially — firms typically skip Stage 2 and jump directly from Stage 1 to Stage 3 3
- The stages are right but the barriers between stages are different from what I'd expect 0
- No — the most common progression I observe is (Provide response in below) 0
- Other 1



47. What are the **top three real-world barriers** preventing Tier 1 cell manufacturers from advancing to Stage 2 or Stage 3 predictive risk tools? (Select up to three)

- Data unavailability — internal data not systematically collected 8
- Data quality — existing data is inconsistent or unreliable 6
- Technical capability gap — no in-house ML / data science function 2
- Budget constraints — capital not allocated to risk analytics 5
- Organisational resistance — risk function not trusted with predictive outputs 2
- Regulatory uncertainty — unclear if tool outputs meet compliance requirements 0
- Vendor dependency — lack of open-source / affordable tools 0
- Leadership priority — SRM not a board-level strategic priority 3
- Other 1



48. The research proposes these **investment thresholds** for each adoption stage. Do these figures reflect realistic implementation costs for a mid-scale Tier 1 cell manufacturer? (If any figure is off, your suggested range: Stage 1: EUR _ | Stage 2: EUR _ | Stage 3: EUR _

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Responses

Latest Responses

...

49. What is a realistic **implementation timeline** for each stage from a standing start (no prior predictive tool capability)?
(If any figure is off, your suggested range: Stage 1: EUR _ | Stage 2: EUR _ | Stage 3: EUR _

3

Responses

Latest Responses

...

50. For **Tier 3 raw material suppliers**, which single barrier to SRM improvement do you consider most fundamental — the barrier that, if re moved, would unlock the most progress?

